

motor system is slightly slower and less expensive but still provides plentiful power and runtimes, the brushless motor system offers more power, requires less maintenance and is more efficient.

Drive system. In the drive system, usually the electric motor transfers its power to the propeller via some sort of drive shaft. The drive shaft has some sort of tube (stuffing tube) that helps it to exit the hull and establish a rigid connection with the propeller mount.



Fig. 3: Near-finished DIY kit of the RC boat without the top shell and water-tight covers

Radio system. The radio system is a combination of the radio frequency (RF) transmitter and receiver. Now, the good old 27MHz (and 75MHz) radio systems have, for the most part, been replaced with 2.4GHz radio systems. 2.4GHz radio systems are less prone to interference from other radio sources and free from undesired frequency conflicts.

Battery. An RC electric boat would need a suitable onboard battery pack. There are several things to consider when purchasing a battery pack, and one of the important things to consider is the type of battery. Ni-MH batteries are less expensive and provide solid performance. Ni-MH packs have a sharper discharge curve, which means, the speed and performance difference between the start and end of a run tends to be greater.

Lithium-polymer batteries are lighter in weight than Ni-MH packs

and have a flatter discharge curve, which means, the performance from start to end is more consistent.

Further, lithium-polymer batteries are free from inherent performance degradation problems that Ni-MH packs have. First consideration with a lithium-polymer battery pack is the C rating, which refers to the amperage-discharge capability of a particular battery. A higher-rating battery pack can handle more load without any issue.

Battery charger. From the battery pack for handheld starters to the receiver pack, having a good battery charger can make it easier to run the boat. Although timer chargers and peak detection chargers are available for RC boats, the latter is recommended for better results.

A timer charger will save you money but a peak-detection charger, also known as peak charger, costs a little more and features advanced charging algorithms to ensure that the battery pack is brought to a complete charge safely.

ESC. Many of the latest RC boat model kits include a programmable ESC. The programming feature allows alteration of the cutoff voltage of the ESC to protect the battery pack from over-discharging. The enhanced option to programme the ESC through the use of an ESC programming module (EPM) will also make it possible to activate braking, set cutoff preferences and choose throttle curves.

DIY tips

These days it is easy to buy DIY kits for RC boats from many online vendors. Independent spare parts for these boats are also available at affordable rates. If you want to drive a boat but you do not have access to a wide area of open water, a mini boat kit is an option. It is smaller than other boats but features similar handling and performance characteristics of its larger counterparts.

These mini boat DIY kits require very little preparation time to get on the water and provide an inexpensive



Fig. 4: Brushed electric motor, propeller shaft and servo



Fig. 5: Propeller, battery pack and electronic speed control (ESC)